

Photosynthesis: How Plants Make Food

Photosynthesis is the process by which green plants, algae and some bacteria convert light energy into chemical energy stored in glucose. It takes place mainly in the chloroplasts of leaf cells, which contain the green pigment chlorophyll.

The overall equation is: six molecules of carbon dioxide plus six molecules of water, using light energy, produce one molecule of glucose and six molecules of oxygen.

Why it matters

Almost all life on Earth depends on photosynthesis. It supplies the oxygen we breathe and forms the base of nearly every food chain. Fossil fuels are ancient stored products of photosynthesis.

The light-dependent reactions

The light-dependent reactions happen in the thylakoid membranes. Chlorophyll absorbs light, which excites electrons. Water molecules are split in a process called photolysis, releasing oxygen as a by-product. The energy of the excited electrons is used to make ATP and NADPH.

The Calvin cycle

The Calvin cycle, also called the light-independent reactions, takes place in the stroma. The enzyme RuBisCO fixes carbon dioxide onto ribulose biphosphate. Using ATP and NADPH from the light-dependent reactions, the cycle produces G3P, a three-carbon sugar used to build glucose.

Limiting factors

The rate of photosynthesis is limited by light intensity, carbon dioxide concentration and temperature. Whichever factor is in shortest supply limits the overall rate. Enzymes such as RuBisCO slow down at low temperatures and denature at very high temperatures.

光合成（日本語のまとめ）

光合成とは、植物が光エネルギーを使って二酸化炭素と水からグルコースと酸素を作る過程である。葉緑体の中で行われる。